CS24309 — THEORY (3.0 CR) | 10-PAGE MASTER REVISION | BIT MESRA | NEP SCHEME

Software Engineering — 10-Page Master Quick Revision

High-Yield Formula Sheet, Metric Tables & Top 10 BIT Mesra PYQ Solutions

⚡ 10-Page Master Quick Revision — Software Engineering (CS24309)

⚡ Master Formula, Estimation & Metric Cheat Sheet

Metric / Model	Core Mathematical Formulation / Rule	Key Exam Takeaway
Function Point (FP)	$\mathrm{FP} = \mathrm{UFC} \times \left(0.65 + 0.01 \times \sum_{i=1}^{14} F_i\right)$	Independent of programming language; measures functionality from user perspective.
Basic COCOMO Effort	$E = a \cdot (\mathrm{KLOC})^b, \quad T_{\mathrm{dev}} = c \cdot (E)^d$	Organic ($a = 2.4, b = 1.05$), Semi-Detached ($a = 3.0, b = 1.12$), Embedded ($a = 3.6, b = 1.20$).
PERT Expected Duration	$T_e = \frac{a + 4m + b}{6}, \quad \sigma^2 = \left(\frac{b-a}{6}\right)^2$	Weights most likely estimate (m) 4 times higher than optimistic (a) and pessimistic (b).
Cyclomatic Complexity	$V(G) = E - N + 2P = P + 1 = \text{Enclosed Regions} + 1$	Specifies upper bound on number of test cases needed for branch coverage.
Mutation Score (MS)	$\mathrm{MS} = \frac{D}{T - E} \times 100\%$	Measures test suite adequacy; D is killed mutants, E is equivalent mutants.

🔥 Top 10 High-Yield BIT Mesra Exam Questions & Solutions

Q1. Given a program with 14 edges and 10 nodes, compute its McCabe's Cyclomatic Complexity and state its significance. (6 Marks)

$$V(G) = E - N + 2P = 14 - 10 + 2(1) = \mathbf{6}$$

Significance: The program contains exactly 6 linearly independent execution paths. A minimum of 6 distinct test cases is mathematically necessary and sufficient to guarantee 100% basis path (branch) test coverage.